

Szu-Ju Chen

(724) 807-5424 | szjuchen@gmail.com | szjuchen.github.io | github.com/szjuchen

Education

Carnegie Mellon University, M.S. in Intelligent Information System at SCS Aug. 2026 – Dec. 2027

- **Courses:** Advanced Natural Language Processing, LLM Systems, Introduction of Question Answering

National Taiwan University, B.S. in Computer Science and Information Engineering Sep. 2021 – Jun. 2025

- **Courses:** Machine Learning, Applied Deep Learning, Computer Vision, Information Retrieval and Extraction

- GPA: 4.15/4.3; Dean's List Award (Ranked 1st in Spring 2025 & Fall 2024); Fu Bell Scholarship (Fall 2024 & Fall 2021)

Work Experience

Software Engineer, Google – New Taipei City, Taiwan Aug. 2025 – Aug. 2026

- Delivered Pixel Clock Timer production mechanism and user interface to set up timer with keypads and preset buttons
- Constructed Pixel Calculator end-to-end evaluation dashboard with 15+ automated metric scripts across 2M+ user data
- Implemented in-app parser evaluating AI-generated Python formulas for complex expression handling in Pixel Calculator

Machine Learning Scientist Intern, Appier – Taipei City, Taiwan Sep. 2024 – May 2025

- Resolved one-epoch overfitting issues in CTR prediction architectures, boosting accuracy by 5%
- Fine-tuned production ranking models and optimized feature engineering pipelines across large-scale advertising data

Software Engineer Intern, Google – New Taipei City, Taiwan Jun. 2024 – Sep. 2024

- Built an Android package for updatable onboarding flows, enabling dynamic feature updates and immediate bug fixes
- Designed modular data transfer workflows and refactored components into reusable libraries; passing 7/7 internal tests

Student Training Engineering Program Intern, Google – New Taipei City, Taiwan Jun. 2023 – Sep. 2023

- Integrated multiple network FTP speed testing features into internal modem testing Android app
- Delivered high-precision network performance analytics under 5G environments, fully adopted by internal teams

Publications

Retrieving the Right Law: Enhancing Legal Search with Style Translation, SIGIR2025 Short Paper

Szu-Ju Chen, Jing Jin, Sheng-Lun Wei, Chien-Hung Chen, Hsin-Hsi Chen | 10.1145/3726302.3730246

- Designed style translation module and retrieval algorithms for legal question answering using related Taiwanese laws
- Trained a Bi-Encoder model and fine-tuned Taiwan-LLaMA-3B using original user queries and LLM-rewritten queries

Research Experience

MACHI Program in DATA Lab, Dr. Xia Ben Hu – Rice University, United States Feb. 2025 – Mar. 2025

- Applied dynamic routing algorithms to multi-agent visual reasoning, cutting latency by 50% and raising accuracy by 7%
- Awarded 2nd Place out of all participants in the 2025 Rice MACHI Poster Competition

Information Retrieval Lab, Dr. Pu Jen Cheng – National Taiwan University, Taiwan Sep. 2023 – Jan. 2025

- Engineered a pipeline integrating LightGCN trained on millions of user-track data with LLM-generated expansions
- Built a hybrid recommender system for genre and keyword queries, earning the Department Research Excellence Award
- Investigated autoregressive image retrieval utilizing VQ-VAE indexing and contrastive learning representations

Projects

Visual Grounding & Image Inpainting Pipeline Dec. 2024

- Designed a computer vision pipeline integrating visual grounding with image inpainting for indoor object replacement
- Fine-tuned and benchmarked diffusion-based models against PowerPaint and BrushNet on a custom-curated dataset

Konnect – Cross-Platform Fan Interaction App Sep. 2023 - Jun. 2024

- Led development of a full-stack React Native app integrated with Google Maps, GCP backend, and real-time data sync
- Achieved MVP stage and passed initial testing with user posting, activity tracking, and content sharing features

Chinese Grammatical Error Correction with Seq2Seq Models Apr. 2023 - Jun. 2023

- Benchmarked BART, mT5, and Taiwan-LLaMA models to evaluate zero-shot and fine-tuned correction performance
- Finetuned google/mt5-small on open-source corpora, evaluating with self-designed Exact-Match and M^2 metrics

Skills

Programming Languages: C++, C, Python, Java, JavaScript, Kotlin, SQL

Machine Learning: LLMs, recommendation systems, information retrieval, computer vision, RAG, multi-agent

App Development: Android apps, React Native, GCP, Firebase, Git